$Q = It$

Q = charge (C)

I = current (A)

t = time (s)

Potential difference - L 2

A cell makes one end of the circuit positive and the other negative. This sets up a potential difference across the circuit.

- Potential difference is measured using a voltmeter. A voltmeter is always set up in parallel.
- The p.d. is defined as the energy transferred per unit charge. Another meaning of energy transfer is work done

$V = \frac{W}{Q}$

V = potential difference (V)

W = work done (J)

Q = charge (C)

L1 - Electrons and Charge

Electron movement is the essence of electricity.

Electric current is the flow of charge carriers and is measured in units of amperes (A)

- Charge carriers can be either positive or negative.
- Charge can flow between two conductors.

The direction of conventional current in a metal is from positive to negative.

For electricity to flow, there must be a path that is a complete loop.

- In electrical wires, the current is a flow of electrons
- Current is measured using an ammeter

Ammeter should always be connected in series.

- Electric charge is measured in units of coulombs (C)

L3 - Series Connection, Parallel Connection

Series connection is a circuit where the components are connected end-to-end in a line

Parallel circuit is a circuit where all components are connected across each other loads

In series connection: $I_{total} = I_1 = I_2$

$$U_{total} = U_1 + U_2$$

$$R_{total} = R_1 + R_2$$

In parallel connection: $U_{total} = U_1 = U_2$

$$I_{total} = I_1 + I_2$$

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2}$$

L4 - Circuit Symbols, Electromotive Force, Internal Resistance

Switch : Turn the circuit on or off Fixed resistor : A resistor limits the flow of current.

Thermistor : The resistance of a thermistor depends on its temperature Temperature ↑
resistance ↓

Light - dependent resistor (LDR) : The resistance of an LDR depends on the light intensity

Diode : A diode allows current to flow in one direction only.

Light - emitting diode (LED) : It emits light when a current passes through it.

Ammeter : Used to measure the current

Voltmeter : Use to measure the potential difference lost volts = $I \cdot r$ (Ohm's law)

$$E = IR + Ir = I(R + r)$$

E = e.m.f (V)
I = current (A)
R = load resistance (Ω)
r = internal resistance (Ω)

L7 – Kirchhoff's Laws

Kirchhoff's first law: The sum of the currents entering a junction is always equal to the sum of the currents leaving the same junction.

- A junction is a point where at least three circuit path meet.
- A branch is a path connecting two junctions.

$$I = I_1 + I_2 + I_3$$

Electromotive force

The electromotive force (e.m.f) is the amount of energy transferred per coulomb of charge when charge passes through a power supply. e.m.f. is measured in Volts (V)

$$\text{e.m.f} = \frac{\text{energy transferred}}{\text{charge}}$$

• Potential difference is also the energy transferred per unit charge.

$$\underline{\text{potential difference}} = \frac{\text{energy transferred}}{\text{charge}}$$

Internal resistance

All power supplies have some resistance between their terminals. This is called internall resistance. V_R is the terminal potential difference

Terminal potential difference = $I \cdot R$ (Ohm's law)

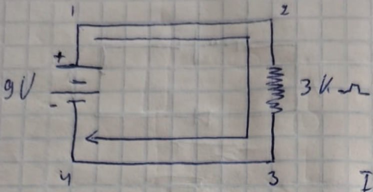
V_r is the lost volts

<u>lost volts</u> = e.m.f - terminal p.d.

L5:

3)

1)



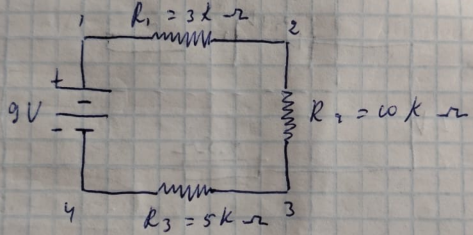
$I = ?$

$V = 9V$

$R = 3k\Omega$

$$I = \frac{V}{R} = \frac{9}{3000} = 3mA$$

2)



$I = ?$

$R_{total} = R_1 + R_2 + R_3 = 3000 + 10000 + 5000 = 18k\Omega$

$V = 9V$

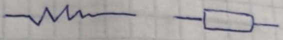
$$I = \frac{9}{18000} = 500\mu A$$

$V_{R_1} = I_{R_1} \cdot R_1 = 500 \cdot 3 = 1.5V$

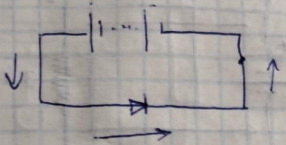
$V_{R_2} = I_{R_2} \cdot R_2 = 500 \cdot 10 = 5V$

$V_{R_3} = I_{R_3} \cdot R_a = 500 \cdot 5 = 2.5V$

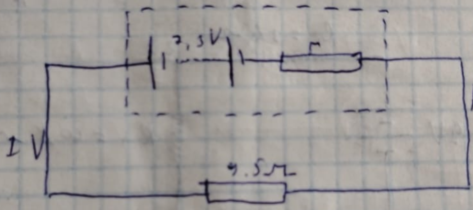
3) Resistors



4) B



5)



$E = 7.3V$

$R = 9.5\Omega$

$r = 0.3\Omega$

a) $I = ?$

b) $V_{lost} = ?$

$$E = I(R + r)$$

a) $I = \dfrac{E}{R+r} = \dfrac{7.3}{9.5+0.3} = 0.745 =$

b) $V_{lost} = Ir = 0.7 \cdot 0,3 = 0.21V$

L1:

1) $I = 8\,mA$

$Q = It$

$I = \frac{Q}{t}$

(B) $I = \frac{4}{500} = 8 \cdot 10^{-3}\,A = 8\,mA$

L2:

1) (A) electrons in the weire are moving totally randomly within the were

(2) $Q = 1C$

$Q = It$

(C) $Q = 5 \cdot 10^{-3} \cdot 200 = 1C$

(3) $I = 10\,mA = 10 \cdot 10^{-3}$

$t = 0.5\,h = 30\,min = 1800\,sec$

(C)

$Q = ?$

$Q = It = 10 \cdot 10^{-3} \cdot 18 \cdot 10^{2} = 18C$

$$V_{out} = \left(\frac{R_{target}}{R_1 + R_2}\right) V_{in}$$

L9 - Voltage Divider Circuits

Ohm's law : $I = \frac{E}{R}$

Series circuit is often called a voltage divider for its ability to proportion - or divide - the total voltage into fractional portions of constant ratio.

Voltage drop across any resistor $E_n = I_n R_n$

Current in a series circuit $I_{total} = \frac{E_{total}}{R_{total}}$

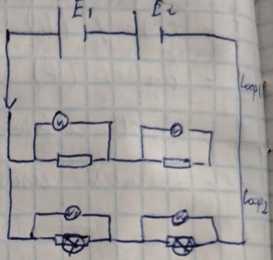
Voltage drop across any series resistor :

$$E_n = \frac{E_{total}}{R_{total}} \cdot R_n \quad \text{or}$$

$$E_n = E_{total} \cdot \frac{R_n}{R_{total}}$$

Kirchhoff's second law: The total e.m.f in a closed circuit is equal to the sum of the potential difference across the components.

$E_1 + E_2 = V_1 + V_2 = V_3 + V_4$



- $Q = ne$
- $I = \frac{Q}{t}$ · $Q = It$ · $V = IR$ · $V = \frac{W}{q}$ (potential difference)
- $E = qV$ · $V = \frac{E}{q}$ · $P = VI$ · $P = I^2R$ · $R = \frac{V_2}{I}$
- $P = \frac{E}{t}$ · $Q = C \cdot V$ · $E = \frac{1}{2}CV^2$

$V_{lost} = I \cdot r$

$V_{lost} = E - V$

E ↓ e.m.p, V ↓ potential difference

- $W = V \cdot Q$

L10 - Potential divider circuit.

When 2 resisters are conneted in series, through Kirchhoff's second law, the potential difference across the power source is divided betwen them.

Potential dividers are circuit which produce an output voltage as a fraction of its input voltage.

Potential dividers have 3 main purpose.

- To provide a variable potential difference
- To enable a specific potential difference to be chosen.
- To split the potential difference of a power source between two or more components

(V~ - un gabagirob ela)

$$V_{out} = \frac{R_2}{R_1 + R_2} \cdot V_{in} \quad \text{or} \quad V = \frac{R_1}{R_1 + R_2} \cdot V_{in}$$

$$V_{in} = \frac{V_{out}}{\frac{R_1}{R_1 + R_2}} = \frac{(R_1 + R_2)}{R_1} \cdot V_{out}$$

L3

1) In a series circuit, voltage drops add to equal (through all component) the total current is equal through all components. resistance add to equal the total power dissipation add to equal the total

2) in a parallel circuit, voltage is equal across all component
currents add to equal the total resistances diminish to equal the total

L4:

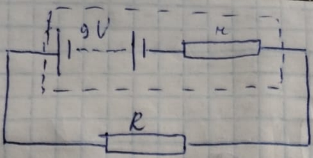
1) B $V = IR$ potential difference = current resistance

2) $R = \frac{V}{I}$ $V = \frac{E}{Q} = \frac{\frac{1}{2}mv^2}{It} = [kg\,m^2 s^{-2}][As]$

$R = \frac{V}{I} = kg \cdot m^2 \cdot s^{-2} \cdot A^{-1} \cdot A^{-1} = kg \cdot m^2 \cdot s^{-3} \cdot A^{-2}$

B

3)



$Q = 240\,C$

$E = 1440\,J$

$t = 120\,sec$

$r = ?$

$Q = It$

$I = \frac{Q}{t} = \frac{240}{120} = 2A$

$P = I^2 R \quad P = \frac{E}{t}$

$\frac{E}{t} = I^2 R \quad R = \frac{E}{I^2 t}$

$R = \frac{1440}{4 \cdot 120} = 3\,\Omega$

$E = I(R + r) = 9V$

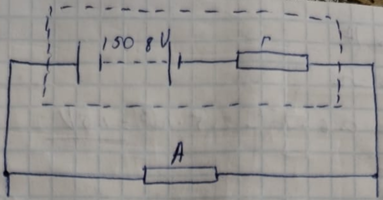
$R + r = \frac{E}{I} = \frac{9}{2} = 4.5\,\Omega$

$R = 3\,\Omega$

$r = 4.5 - 3 = 1{,}5\,\Omega$

A

3)



$V = 1508\ V$

$A = 400\ mA$

$P = 600\ W$

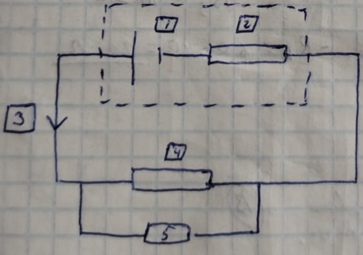
$P = I V \qquad V = \frac{P}{I}$

$V = \frac{600}{0.4} = 1500\ V$

$E = V + I r \Rightarrow r = \frac{E - V}{I}$

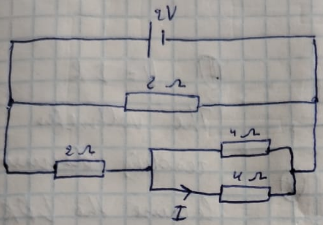
$r = \frac{1508 - 1500}{0,4} = 20\ \Omega$

3)



1 - E
2 - r
3 - I
4 - R
5 - V

3) $E = 2V$



$$\frac{1}{R_{42}} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$R_{1+2} = 2\ \Omega$

$R_{1+2+3} = 2+2 = 4\ \Omega$

$$\frac{1}{R_{total}} = \frac{1}{R_{1+2+3}} = \frac{1}{R_4} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

$R_{total} = \frac{4}{3}\ \Omega$

$$I = \frac{V}{R} = \frac{2}{\frac{4}{3}} = 1{,}5\ A$$

$I = \frac{2}{6} \cdot 1{,}5 = 0{,}5\ A$

$I = \frac{0{,}5}{2} = 0{,}25\ A$

$I = 0.6\ mA$

$R - 2.5k\ \Omega$

$V = IR = 0.6 \cdot 10^{-2} \cdot 2.5 \cdot 10^{3} = \underline{1.5V}$

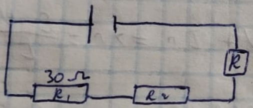
$r = 1,3\ \Omega$

$E = I(R + r)$

$\mathcal{E} = V + Ir = 1.5 + 0,6 \cdot 10^{-3} \cdot 1.3 \cdot 10^{3} = \underline{2.28V}$

L7:

1)



$R_{total} = 60\ \Omega$

$R_1 = 30\ \Omega$

$R_2 = ?$

$R_3 = 10\ \Omega$

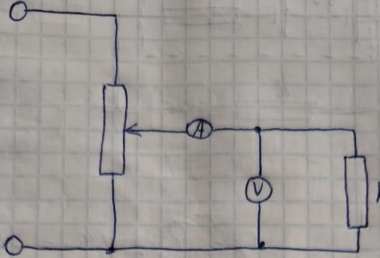
$R_2 = R_{total} - R_1 - R_3 = 60 - 40 = \underline{20\ \Omega}$

L8:

1) $\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2}$ (B)

2) (C)

4)



$V = IR$ $R = \frac{V}{I}$ R is much smaller than R_v

A

L6:

1) $l = 800 m$

$V = 16 V$

$I = 0.6 A$

$R = 0.005 \Omega$

C

$R_t = 800 \cdot 0,005 = 4 \Omega$

both wires $= 4 + 4 = 8 \Omega$

$V = IR = 0.6 \cdot 8 = 4.8 V$

$V_{e.m.f} = 16 + 4.8 = 20.8 V$

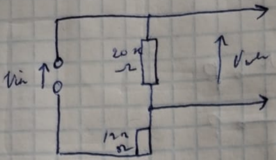
2) Voltage - V, V Resistance - R, Ω

Current - I, A

3)

L 10 :

1)



$V_{out} = 5.3\,V$

$V_{in} = ?$

$R_1 = 20\,k\Omega$

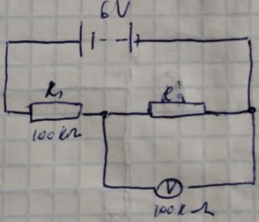
$R_2 = 12\,k\Omega$

$$V_{in} = \frac{\frac{1}{R_1}}{R_1 + R_2} \cdot V_{out} =$$

$$= \frac{\frac{1}{20 \cdot 10^3}}{32 \cdot 10^3} \cdot 5.32$$

$$= \frac{32}{20} \cdot 5.3 = 8.5\,V$$

2)



$V = 6\,V$

$R_1 = 100\,k\Omega$

$R_2 = 100\,k\Omega$

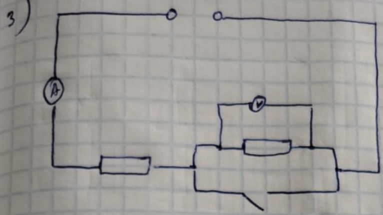
$$V_{out} = \frac{R_2}{R_1 + R_2} \cdot V_{in}$$

$R_{total} = 100 + 50 = 150\,k\Omega$

$I = \frac{6}{150} = 40\,\mu A$

$R_2 = \frac{40}{2} = 20\,\mu A.$

3)

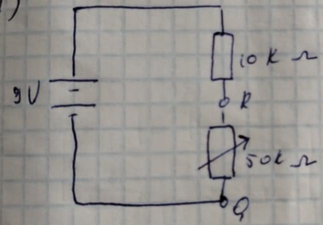


C

When the switch is closed,

I increses, V decreases to zro

4)



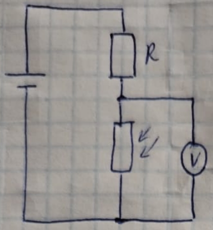
$$V_2 = V_{e.m.f} \cdot \frac{R_2}{R_1 + R_2} = 9 \cdot \frac{50}{60} = 7.5V$$

$$V_2 = \frac{9 \cdot 0}{10} = 0V$$ B

5) B $\frac{V_0 R_2}{R_1 + R_2}$

L9:

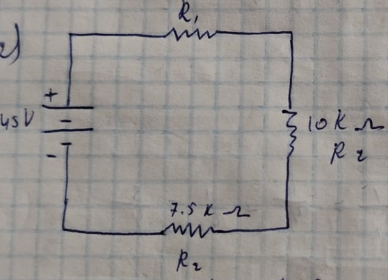
1)



(A) $V = IR$

The voltmeter reading decrease because the LDR resistance decrease

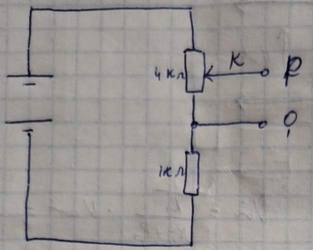
2)



	R_1	R_2	R_3	Total	
E	10	20	15	45	Volts
I	2m	2m	2m	2m	Amps
R	5K	10K	7.5K	22.5K	Ohms

L11:

1)

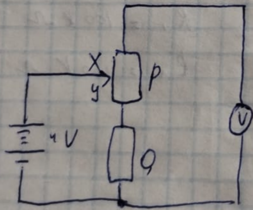


$$V_2 = V_{e.m.f} \cdot \frac{R_1}{R_1+R_2} = \cancel{25}^{5} \cdot \frac{4}{\cancel{5}} = 20V$$

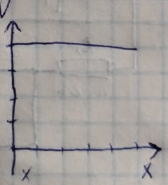
$$V_2 = 25 \cdot \frac{0}{1} = 0V$$

B

2)



A



Because the circuit is series and the voltmeter must be parallel